INCH-POUND

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20 September 1991
SUPERSEDING
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15 January 1971

MILITARY SPECIFICATION

CARTRIDGE, 7.62MM: DIM TRACER, XM276

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers Cartridge, 7.62mm: Dim Tracer, XM276 for use in 7.62mm weapons.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 -Ammunition, Standard Quality Assurance Provisions, General Specifications for

STANDARDS

MILITARY

MIL-STD-109 -Quality Assurance Terms and Definitions
MIL-STD-636 -Visual Inspection Standards for Small Arms Ammunition through Caliber .50

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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MIL-STD-644	-Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1170	-Visual Standards and Comparison Methods for Evaluating Grain Configuration in 7.62mm Cartridge Cases

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect of the date of solicitation.

DRAWINGS

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7643674	-Classification of Cartridge Case Defects
10535780	-Packing, Cartridges, 7.62mm, Linked, Shipping and Storage Container, M548 (functional)
10535781	-Marking, Cartridges, 7.62mm, Linked, Shipping and Storage Container, M548 (functional)
10542714	-Cartridge, 7.62mm, Dim, Tracer, XM276

INSPECTION EQUIPMENT DRAWINGS

LI10542714	-Index of Inspection Equipment List for: Cartridge, 7.62mm, Tracer, XM276

PUBLICATIONS

DEPARTMENT OF DEFENSE

SCATP-7.62	-Ammunition Ballistic Acceptance Test Methods, Test Procedures for 7.62mm Cartridges

(Copies of other government documents, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Non-government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS specified in the solicitation.

AMERICAN SOCIETY OF TESTING AND MATERIALS

E92 -Metallic Materials, Vicker Hardness of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadephia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Cartridge. The cartridge shall comply with all requirements specified on Drawing (Dwg) 1052714, all associated drawings and with all requirements specified in applicable specifications and standards.

3.1.1 Material. All material shall be in accordance with the applicable drawings, specifications and standards.

3.1.2 Parts. All parts shall comply with the applicable drawings, specifications and standards.

3.2 Bullet extraction. The force required to extract the bullet from the cartridge case shall not be less than 60 pounds.

3.3 Residual stress. The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Waterproof. The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 7.5 psi for 30 seconds.

3.5 Accuracy. The average of the mean radii of all targets of the sample cartridges, fired in standard accuracy test weapons over a range of 600 yards, shall not exceed 15.0 inches.

3.6 Trace. When viewed at night from behind and above or to one side of the gunner, the bullet of the tracer cartridge shall exhibit a detectable trace, when viewed with the aid of a starlight scope, from the muzzle of the weapon to a point not less

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than 850 yards from the muzzle. The trace shall be also at a very low level of luminosity and preferably invisible when viewed with the naked eye.

3.7 Action time. The action time (overall primer ignition, propellant burning and bullet-barrel time) of the sample cartridges when conditioned for not less than 2 hours at 68-72 degrees F (°F+), shall not exceed 4 milliseconds.

3.8 Temperature test. The chamber pressure, velocity and function and casualty requirements of the sample cartridges subjected to each of the following storage conditions shall be as noted in the applicable subparagraphs:

a. Conditioned at 70°F for not less than 2 hours and fired at that temperature.

b. Conditioned at 125°F for 4 to 12 hours and fired at that temperature.

c. Conditioned at -65°F for not less than 6 hours and fired at that temperature.

3.8.1 Chamber pressure. The chamber pressure measurement shall be performed by either the copper-crush cylinder or EPVAT (Electronic Pressure Velocity Action Time) test method. However, the test method employed shall be recorded on the test results.

3.8.1.1 Chamber pressure measurement at 70°F by copper-crush cylinder test method. The average chamber pressure of the sample cartridges conditioned at 70°F shall not exceed 50,000 pounds per square inch (psi). The average chamber pressure plus three standard deviations of chamber pressure shall not exceed 55,000 psi. The chamber pressure of an individual sample cartridge shall not exceed 55,000 psi.

3.8.1.2 Chamber pressure measurement at 70°F by EPVAT test method. The average chamber pressure of the sample cartridges conditioned at 70°F shall not exceed 365 Mega Pascals (MPa) (52,940 psi). The average chamber pressure plus three standard deviations of the chamber pressure shall not exceed 400 MPa (58,016 psi). The chamber pressure of an individual sample cartridge shall not exceed 400 MPa (58,016 psi).

3.8.1.3 Chamber pressure measurement at 125°F and -65°F by copper-crush test method. The average chamber pressure of the sample cartridges subjected to 125°F and -65°F storage conditions shall not exceed 55,000 psi nor vary by more than plus (+) 7,500 psi or minus (-) 15,000 psi from the average chamber pressure of the sample cartridges of the same lot tested for the requirements of 3.8.1.1.

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3.8.1.4 Chamber pressure measurement at 125°F and -65°F by EPVAT test method. The average chamber pressure of the sample cartridges subjected to 125°F and -65°F storage conditions shall not exceed 400 MPa (58,016 psi) nor vary by more than + 50 MPa (7,252 psi) or - 100 MPa (14,504 psi) from the average chamber pressure of the sample cartridges of the same lot tested for the requirements of 3.8.1.2.

3.8.1.5 Velocity. The average velocity of the sample cartridges, conditioned at 70°F shall be 2,750 feet per second (ft/sec) ± 30 ft/sec for gilding metal clad steel jacketed bullets or 2,680 ft/sec ± 30 ft/sec for gilding metal jacketed bullets at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 32 ft/sec.

3.8.1.6 Velocity. The average velocity of the sample cartridge subjected to 125°F and -65°F storage conditions shall not vary from the average velocity of the sample cartridges of the same lot conditioned at 70°F by more than + 150 or - 250 ft/sec.

3.8.1.7 Function and casualty. The sample cartridges subjected to 70°F, 125°F and -65°F shall function without casualty.

3.9 Bullet integrity. The bullet of the cartridge shall not burst either in its passage through the barrel or in slight; neither shall the jacket of the bullet or any part thereof strip from the slug when the cartridge is fired.

3.10 Port pressure. The port pressure measurement can be performed by either the copper-crush cylinder or EPVAT test method. The average port pressure of the sample cartridges when conditioned for not less than 2 hours at 68-70°F and fired at that temperature shall be as noted in the applicable sub-paragraphs:

3.10.1 Measurement of copper-crush cylinder test method. The average port pressure of the sample cartridges shall be 12,500 psi ± 2,000 psi.

3.10.2 Measurement by EPVAT test method. The average port pressure of the sample cartridges shall be not greater than 85 MPa (12,328 psi) and shall not be less than 60 MPa (8,702 psi).

3.11 Grain configuration. The grain configuration of the sidewall of the finished cartridge case shall fall within the range defined by the grain configuration standards illustrated in MIL-STD-1170, Figures 1 through 6.

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3.12 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following. The metal parts of the cartridge shall be free of cracks, splits, perforations, burrs and foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by any cleaning agent. All parts and assemblies shall be fabricated, loaded and assembled in a thorough workmanlike manner. In addition, the cartridge shall comply with the standards specified in the 7.62mm section of MIL-STD-636. Extreme care should be exercised to avoid contamination of primers or propellant by oil, grease or other foreign matter.

3.13. First article inspection. Requirements for submission of the First Article Inspection shall be as specified in the contract. When specified, a sample shall be subjected to First Article Inspection (see 6.2) in accordance with 4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

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b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply. The first article sample shall consist of the following items in sample quantities as indicated below:

Part Description	Drawing	Quantity
Cartridge, 7.62mm Dim Tracer, XM276	10542714	4,440
Filler, point	7553760	5/each final station/each machine
Jacket	7553759	5/each final station/each machine
Jacket (alternate)	10523993	5/each final station/each machine
Bullet	10542715	5/each final station/each machine
Bullet (alternate)	12944379	5/each final station/each machine
Case	10521997	5/each final station/each machine

4.3.2 Inspections to be performed. See MIL-A-48078 and Tables I and II specified herein.

4.3.3 Rejection. See MIL-A-48078 and Tables I and II specified herein.

4.3.3.1 Function and casualty. The sample shall be rejected if any defect from function and casualty testing plus firing defects observed in all other firing tests exceeds the acceptance numbers specified in Table II.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50704A(AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
	Cartridge, 7.62mm, XM276, Dim Tracer		1 OF 5	10542714
				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Filler, point (Dwg. 7553760) Examination for defects	5 1/	Acc-0 Rej-1	3.1	2/
	Jacket (Dwg. 7553759) Examination for defects	5 1/	Acc-0 Rej-1	3.1	2/
	Jacket alternate (Dwg. 10523993) Examination for defects	5 1/	Acc-0 Rej-1	3.1	2/
	Bullet (Dwg. 10542715) Examination for defects	5 1/	Acc-0 Rej-1	3.1	2/
	Bullet alternate (Dwg. 12944379) Examination for defects	5 1/	Acc-0 Rej-1	3.1	2/
	Case (Dwg. 10521997-1) Examination for defects	5 1/	Acc-0 Rej-1	3.1	2/
	Cartridge, 7.62mm, Dim Tracer, XM276 (Dwg. 10542714)		Acc/Rej	3.1	Table
	Examiantion for Defects: Critical defect	4440	0 1	3.12	
	Major defect	500	2 3		3/
	Minor defect	500	8 9		3/

NOTES:
1/ Each final station, each machine
2/ The final sample units shall be inspected with standard measuring and test equipment for all mandatory requirements specified on the applicable drawing.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER
	Cartridge, 7.62mm, XM276, Dim Tracer				10542714
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Bullet extraction	80	Acc-0 Rej-1	3.2	4.5.1
	Residual stress	80	Acc-0 Rej-1	3.3	4.5.2
	Action at 70°F	80	Acc-0 Rej-1	3.7	4.5.6
	Velocity				
	70°F	40	4/	3.8.1.5	4.5.7, 4.5.9.1
	+125°F	40	4/	3.8.1.6	4.5.7, 4.5.9.3
	-65°F	40	4/	3.8.1.6	4.5.7, 4.5.9.2
	Chamber pressure				
	70°F	40	4/	3.8.1.1/ 3.8.1.2	4.5.8, 4.5.9.1
	+125°F	40	4/	3.8.1.3/ 3.8.1.4	4.5.8, 4.5.9.3
	-65°F	40	4/	3.8.1.3/ 3.8.1.4	4.5.8, 4.5.9.2
	Port pressure				
	70°F	40	4/	3.10	4.5.12, 4.5.9.1

NOTES:

3/ One random sample of 500 cartridges for major & minor defects inspection taken from 4400 cartridges.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			SHEET 3 OF 5	DRAWING NUMBER
	Cartridge, 7.62mm, XM276, Dim Tracer				10542714
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Function and Casualty - Gun Machine, M60				
	70°F	200	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	200	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	200	5/	3.8.1.7	4.5.10, 4.5.9.2
	Function and Casualty - Gun Machine, M134				
	70°F	200	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	200	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	200	5/	3.8.1.7	4.5.10, 4.5.9.2
	Function and Casualty - Gun Machine, M240				
	70°F	200	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	200	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	200	5/	3.8.1.7	4.5.10, 4.5.9.2

NOTES: 4/ The first article sample shall be rejected if the sample fails to comply with the applicable requirement.
5/ The first article sample shall be rejected when function & casualty defects plus firing defects observed on all other firing tests exceed any acceptance numbers listed in Table II.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 5		DRAWING NUMBER
	Cartridge, 7.62mm, XM276, Dim Tracer				10542714
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Function and Casualty - Gun Machine, NF1				
	70°F	100	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	100	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	100	5/	3.8.1.7	4.5.10, 4.5.9.2
	Function and Casualty - Rifle, M14				
	70°F	160	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	120	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	120	5/	3.8.1.7	4.5.10, 4.5.9.2
	Function and Casualty - Rifle, (LAR), FN				
	70°F	160	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	120	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	120	5/	3.8.1.7	4.5.10, 4.5.9.2
	Function and Casualty - Rifle, G3A2				
	70°F	160	5/	3.8.1.7	4.5.10, 4.5.9.1
	+125°F	120	5/	3.8.1.7	4.5.10, 4.5.9.3
	-65°F	120	5/	3.8.1.7	4.5.10, 4.5.9.2
	Waterproof	80	Acc-5 Rej-6	3.4	4.5.3
	Accuracy	180	4/	3.5	4.5.4

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50704A(AR)

PARAGRAPH	TITLE		SHEET 5 OF 5		DRAWING NUMBER
	Cartridge, 7.62mm, XM276, Dim Tracer				10542714
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Trace				
	Machine Gun, M60	150	Acc-20 Rej-21	3.6	4.5.5
	Machine Gun, Test, T65E1	150	Acc-20 Rej-21	3.6	4.5.5
	Bullet Integrity	6/	Acc-0 Rej-1	3.9	4.5.11
	Grain configuration	20	Acc-0 Rej-1	3.11	4.5.13
	Hardness				
	Case Head	20	Acc-0 Rej-1	3.1	4.5.14.2
	Case Sidewall	20	Acc-0 Rej-1 4/	3.1	4.5.14.1

NOTES:

6/ This shall be performed concurrently with trace testing.

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TABLE II

First Article Acceptance and Rejection Numbers for Firing Defects

Defects	Accept	Reject
1. Misfire 1/		
a. No vent hole, or obstruction in vent area	0	1
b. Other	0	1
2. Bullet remaining in bore	0	1
3. Hangfire 2/	0	1
4. Primer Leak		
a. Perforation in firing pin, indent in primer cup		
(1) Machine guns (other than M60, M240, M134)	5	6
(2) Rifles (other than M14)	1	2
(3) Machine gun M60, M240, M134	0	1
(4) Rifle, M14	0	1
b. Escape of gas through primer cup other than 4a.	1	2
c. Escape of gas around primer cup	76	77
d. Blown primer	0	1
e. Primer falls out of pocket on retraction of bolt	0	1
f. Primer remains in pocket but is physically loose.	1	2
5. Case casualties		
a. Longitudinal split 3/		
(1) Neck and shoulder (I or S)	51	52
(2) Body (J)	3	4
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through Head (M)	0	1

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TABLE II
(continued)

First Article Acceptance and Rejection
Numbers for Firing Defects

Defects	Accept	Reject
5. Case casualties		
b. Circumfrential rupture 3/		
(1) Partial, shoulder of body (J and S)	0	1
(2) Partial, body (K)	0	1
(3) Partial Head (L)	0	1
(4) Complete	0	1
6. Failure to extract	0	1
7. Weapon stoppage 4/	0	1
Total of defects 4a(1), 4a(2) 4c, & 5a(1)	77	78
Total of all other listed defects	7	8

1/ Each cartridge that misfires shall be disassembled and examined for the presence of the vent hole in the primer pocket or any obstruction in vent hole area of the primer pocket that can be assignable as the cause for the misfire. If the vent hole is missing or obstructed, the first article quantity shall be rejected.

2/ For the definition of hangfire, see SCATP-7.62, 10.6.2.

3/ For the location of defects indicated by letters in parentheses, see Dwg. 7643674.

4/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture and failure to extract, observed in all tests shall be included.

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4.4 Quality conformance inspection.

4.4.1 Lot formation. In accordance with MIL-A-48078.

4.4.1.1 Component parts. Unless otherwise specified, component parts shall be homogeneous and of a size convenient to the contractor and inspected, tested and accepted by the contractor. The cartridge lot shall contain:

a. Cartridge cases from one unchanged process and from one manufacturer.

b. Bullets from one unchanged process and from one manufacturer.

c. Primers from one lot interfix number and from one manufacturer.

d. Propellant from no more than two lot numbers and from one manufacturer.

4.4.1.2 Lot identification. Each lot of ammunition shall be identified as to type, caliber and model, lot number in accordance with MIL-STD-1168 and the supplier's identification as assigned by the procuring activity.

4.4.2 Examination for defects.

a. Major and minor defects. Examination for major and minor defects shall be performed for each of the major and minor defects listed in Table III using sampling plans and acceptance criteria in accordance with Table IV. Inspection Level II shall apply to major defects and Inspection Level IV shall apply to minor defects except Level V shall apply to workmanship. All nonconforming cartridges shall be rejected.

b. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects (see Table III). If a visual critical defect is found in a sample either just prior to a firing test or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot that has been packed or is in the process of being packed, shall be rejected. In addition, the portion of the lot remaining to be packed shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects.

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TABLE III
QUALITY CONFORMANCE CLASSIFICATION OF DEFECTS

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	CARTRIDGE				
1.	Discolored, dirty, oil, smeared			X	
2.	Corroded or stained, if etched		X		
3.	Mixed ammunition types (see 4.4.4)	X	X		
3a.	Missing or partially missing paint identification (see MIL-STD-636)				X
	CASE				
4.	Round Head		X		
5.	Dent				X
6.	Split case				
	in K, L or M location	X			
	In I, S or J location	X			
7.	Perforated case	X			
8.	Draw scratch				X
9.	Scratch			X	
10.	Beveled underside not crimped in cannelure		X		
11.	Case mouth not crimped in cannelure		X		
12.	Scaly metal				X
13.	No chamfer on head (rim)		X		
14.	Fold			X	
15.	Wrinkle			X	
16.	Buckle			X	
17.	Bulge			X	
18.	Illegible or missing head stamp			X	
19.	Defective head			X	
20.	Defective mouth			X	
21.	No visible evidence of mouth anneal		X		

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TABLE III
(continued)
QUALITY CONFORMANCE CLASSIFICATION OF DEFECTS

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	BULLET				
22.	Dent			X	
23.	Scratch			X	
24.	Split bullet jacket		X		
25.	Loose bullet		X		
26.	Missing cannelure		X		
27.	Scaly metal (bullet)				X
28.	Upset (crooked) point			X	
29.	Exposed steel (clad jacket)			X	
30.	Blunt point			X	
31.	Defective cannelure			X	
	PRIMER				
32.	No primer	X			
33.	Cocked primer	X			
34.	Inverted primer	X			
35.	Loose primer		X		
36.	Nicked or dented primer			X	
37.	No waterproofing material (primer pocket joint			X	
38.	Defective crimp			X	
	CARTRIDGE-GAGING				
39.	Total Length		X		
40.	Cartridge profile failure (requiring more than 20 lbs dead weight to insert in profile and alignment gage)		X		
41.	Diameter of extractor groove, max		X		
42.	Diameter of extractor groove, min			X	
43.	Diameter of head		X		
44.	Thickness of head		X		
45.	Length of shoulder datum		X		
46.	Depth of primer		X		

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TABLE III
(continued)
QUALITY CONFORMANCE CLASSIFICATION OF DEFECTS

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	CARTRIDGE-WEIGHING				
47.	Weight, min 2/	X			
	CARTRIDGE-OTHER				
48.	Workmanship (see 3.12 and 4.4.2.a)			X	

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 38.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 25 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 25 grains of propellant shall be classed as a critical defect.

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TABLE IV.
ATTRIBUTES SAMPLING PLAN

Lot Size	Inspection Level I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; Asterisk indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

c. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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4.4.3 Tests. The tests listed in Table V shall be conducted in accordance with the methods of inspection specified in 4.5.

4.4.3.1 Test samples. The quantities for the various tests shall be as specified in Table V. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

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TABLE V

Quality Conformance Tests

	Ambient Temperature	70°F 4.5.10.1	-65°F 4.5.10.2	+125°F 4.5.10.3	Requirement Paragraph
Bullet extraction 1/	25				3.2
Residual Stress (mercurous nitrate) 1/	50				3.3
Waterproof 2/	50				3.4
Accuracy 3/	90				3.5
Trace 4/					3.6
Gun, machine, 7.62mm, M60	100				
Gun, machine, 7.62mm, T65E1	100				
Bullet integrity 5/	--	--	--	--	3.9
Grain configuration 1/	10				3.11
Hardness:					
Head 6/	10				3.1
Sidewall 3/	10				3.1
Function & Casualty 7/					3.8.1.7
Gun, machine M60		150	150	150	
Gun, machine M240		150	150	150	
Gun, automatic M134		200	200	200	
Rifle, M14		120	80	80	
Rifle, (LAR) FN		120	80	80	
Rifle, G3A2		120	80	80	
Gun, machine NF1		100	100	100	

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TABLE V
(continued)
Quality Conformance Tests

	70°F 4.5.10.1	-65°F 4.5.10.2	+125°F 4.5.10.3	Requirement Paragraph
Measurement by copper-crush cylinder				
Action time 1/	50			3.7
Velocity 8/	20	20	20	3.8.1.5, 3.8.1.6
Chamber pressure 8/	20	20	20	3.8.1.1, 3.8.1.3
Port pressure 3/	20			3.10.1
Measurement by EPVAT 9/	30	30	30	
Action time 1/	--			3.7
Velocity 8/	--	--	--	3.8.1.5, 3.8.1.6
Chamber pressure 8/	--	--	--	3.8.1.2, 3.8.1.4
Port pressure 3/	--			3.10.2

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1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of ten or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three but less than ten cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirements.

3/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

4/ The lot shall be rejected if more than twenty shots in either weapon fail to comply with the applicable requirement. No second sample permitted.

5/ This test shall be performed in conjunction with the "Trace Test". Failure of any cartridge to comply with the applicable requirement shall be cause for rejection of the lot. No second sample permitted.

6/ Failure of one or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. No second sample permitted.

7/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table VI. If the number of defects found in the first test exceeds the acceptance number for the first sample, but is equal; to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all the service weapons specified therefore. This procedure shall apply regardless of the weapon or weapons in which firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

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8/ Failure of the cartridges in any sample to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test for the temperature or temperatures at which the failure occurred. Failure of the cartridges of the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

9/ EPVAT test method for chamber pressure, port pressure, velocity and action time shall be performed simultaneously at all temperatures except for action time and port pressure which are performed only at 70°F ± 2°F.

4.4.3.2 Firing defects. Firing defects and acceptance numbers shall be as specified in Table VI.

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TABLE VI

Quality Conformance Acceptance Numbers for Firing Defects

Defects	ACCEPTANCE NUMBER COPPER-CRUSH CYL First Sample 1/	ACCEPTANCE NUMBER COPPER-CRUSH CYL Cumulative (1st & 2nd Sample) 1/	ACCEPTANCE NUMBER EPVAT First Sample 1/	ACCEPTANCE NUMBER EPVAT Cumulative (1st & 2nd Sample) 1/
1. Misfire 2/				
a. No vent hole, or obstruction in vent area 3/	0	-	0	-
b. Other	1	2	1	2
2. Bullet remaining in bore 3/	0	-	0	-
3. Hangfire 4/ 3/	0	-	0	-
4. Primer leak				
a. Perforation in firing pin indent in primer cup				
(1) Machine guns (other than M60, M240, and M134)	6	14	6	14
(2) Rifles (other than M14)	2	4	2	4
(3) Machine gun M62, M240 and M134 5/	0	-	0	-
(4) Rifle, M15 5/	0	-	0	-
b. Escape of gas through primer cup other than 4a.	1	2	1	2
c. Escape of gas around primer cup 6/	80(64)	189(142)	77(61)	186(139)
d. Blown primer	0	1	0	1
e. Primer falls out of pocket on retraction of bolt	0	1	0	1
f. Primer remains in pocket but is physically loose	2 (1)	5 (2)	2 (1)	5 (2)

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TABLE VI

Quality Conformance Acceptance Numbers for Firing Defects

Defects	ACCEPTANCE NUMBER COPPER-CRUSH CYL First Sample 1/	ACCEPTANCE NUMBER COPPER-CRUSH CYL Cumulative (1st & 2nd Sample) 1/	ACCEPTANCE NUMBER EPVAT First Sample 1/	ACCEPTANCE NUMBER EPVAT Cumulative (1st & 2nd Sample) 1/
5. Case casualties				
a. Longitudinal split 7/				
(1) Neck & shoulder (I & S)	54 (43)	132 (99)	52 (41)	130 (97)
(2) Body (J)	4 (3)	8 (6)	4 (3)	8 (6)
(3) Body (K)	1	2	1	2 (4)
To head (L)	0	1	0	1 (5)
Through head (M)	0	1	0	1
b. Circumferential rupture 7/				
(1) Partial, shoulder of body (J & S)	1	2	1	2
(2) Partial body (K)	0	1	0	1
(3) Partial head (L)	0	1	0	1
(4) Complete	0	1	0	1
6. Failure to extract	0	1	0	1
7. Weapon stoppage 8/	0	1	0	1
Total of defects 4a(1), 4a(2), 4c, & 5a(1)	81(64)	190(146)	78(61)	187(143)
Total of all other listed defects	8(6)	17(13)	8(6)	17(13)

1/ Numbers in parentheses indicate acceptance numbers to be applied during reduced testing when function and casualty testing is conducted only in the M60 machine gun, M240 machine gun, 134 automatic gun and M14 rifle.

2/ Each cartridge that misfires shall be disassembled and examined for presence of vent hole in primer pocket or any obstruction in the vent hole area of the primer pocket that can be assignable as the cause for misfire. If the vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted.

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3/ No second sample permitted. Lot shall be rejected.

4/ For the definition of a hangfire, see SCATP-7.62, 10.6.2.

5/ If one or more defects are found in the first sample, a second sample shall be fired in the weapon(s) in which the defect of cartridges specified under function and casualty of Table V for such weapon(s). Prior to the testing of the second sample, the firing pin of the specified weapon(s) in which the defect originally occurred shall be replaced by a new firing pin. If an additional primer perforation is found in the second sample, the lot shall be rejected.

6/ The total number of cases in which the escape of gas is 50% or more around the periphery of the primer cup shall not exceed 25 for the first sample nor more than 57 for the cumulative sample. When note 1/ applies, the total number of cases in which the escape of gas is 50% more around the primer cup periphery shall not exceed 20 for the first sample nor more than 43 for the cumulative sample.

7/ For location of defects indicated by letters in parentheses, see Dwg. 7643674.

8/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

4.4.4 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy, blank or grenade cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing. In addition, after linking, clipping or prior to insertion into cartons, the cartridges shall be inspected for paint identification missing in excess of 40% of original coverage (Major or Minor defect). This inspection shall be in accordance with 4.4.2.

4.4.5 Inspection equipment. The inspection equipment required to perform the inspections and test prescribed herein, is described in the Index of Inspection Equipment List LI-10542714. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

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4.5 Methods of Inspection.

4.5.1 Bullet extraction. The cartridge shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall not be less than three nor more than six inches per minute. The test shall be conducted in accordance with SCATP-7.62.

4.5.2 Residual stress (mercurous nitrate). The test shall be conducted in accordance with SCATP-7.62.

4.5.3 Waterproof. The test shall be conducted in accordance with SCATP-7.62. The container shall be evacuated to a pressure of 7.5 psi (15 inches of mercury) below atmospheric pressure, and held at that pressure for 30 seconds.

4.5.4 Accuracy. The test shall be conducted in accordance with SCATP-7.62, utilizing the longest range available up to 600 yards. Ranges shorter than 200 yards shall not be used. When accuracy testing is conducted on ranges shorter than 600 yards, the average mean radii of the targets shall be multiplied by the applicable conversion factor given below:

Range (yards)	200	300	400	500
Factor	3.15	2.08	1.57	1.25

4.5.5 Trace. The test shall be conducted in accordance with Appendix A. The cartridges shall be fired in Guns, Machine, 7.62mm, M60 and T6E1. One hundred rounds shall be fired in each gun.

4.5.6 Action time. The test shall be conducted in accordance with SCATP-7.62.

4.5.7 Velocity. The test shall be conducted in accordance with SCATP-7.62.

4.5.8 Chamber pressure. The test shall be conducted in accordance with SCATP-7.62.

4.5.9 Temperature tests. The test shall be conducted in accordance with SCATP-7.62. The weapon(s) in which these tests are fired shall be at room temperature. Prior to firing, reference cartridges shall be conditioned at 68-72°F for not less than two hours and fired at that temperature.

4.5.9.1 70°F test. The test samples shall be conditioned at 70°F ± 2°F for not less than two hours and shall be fired at that temperature.

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4.5.9.2 -65°F test. The test samples shall be conditioned at -65°F ± 5°F for not less than six hours and shall be fired at that temperature.

4.5.9.3 +125°F test. The test samples shall be conditioned at +125°F ± 2°F for 4 to 12 hours and shall be fired at that temperature.

4.5.10 Function and casualty test. The test shall be conducted in accordance with SCATP-7.62 and 4.5.11.1. One weapon of each type shall be used. 4.5.10.1

Gun, automatic, M134. The gun automatic, M134 shall maintain a cyclic rate of 6000 rounds per minute. Barrels shall be replaced after firing 60,000 cartridges (10,000 cartridges per barrel). Firing shall be in 200 round bursts, with cooling to ambient temperature between bursts.

4.5.11 Bullet integrity. The test shall be conducted in accordance with Appendix B. Any bullet recorded as a "Blind" by down range observers either by absence of trace or absence of noise of passing bullet and accompanied by either detection of an abnormally loud report and bright flash by the observer at the gun position or by the presence of any irregularly shaped hole in the screen, shall be considered as a bursting or stripped bullet.

4.5.12 Port pressure. The test shall be conducted in accordance with SCATP-7.62.

4.5.13 Grain configuration. The test samples shall be prepared and evaluated in accordance with MIL-STD-1170.

4.5.14 Hardness testing. The bullets shall be extracted, propellant removed and the primer extracted. Each cartridge case of the sample shall be prepared and placed on the appropriate test fixture for testing in accordance with ASTM-E92.

4.5.14.1 Case sidewall. The average of the hardness values of the sample cases for each prescribed point along the sidewall exterior surface shall be computed and charted in accordance with the drawing requirements.

4.5.14.2 Case head. The individual hardness value for each prescribed point on the head section of each sample case shall be recorded. Any value failing to meet the drawing requirement at a prescribed point(s) shall be cause for remeasurement of hardness at the corresponding point(s) on the opposite side of the primer pocket of the same head section from which the initial value was obtained. The higher of the two measurements shall be recorded as the value of record for determination of conformance to drawing requirements.

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4.5.15 Defect penalty. In any ballistic test except function and casualty, in which the occurrence of a firing defect listed in Table VI prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table VI.

5. PACKAGING

5.1 Level A.

5.1.1 Packing. The cartridges shall be packed in accordance with applicable drawing 10535780.

5.1.2 Marking. Marking shall be in accordance with applicable drawing 10535781.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory)

6.1 Intended use. The cartridges covered by this specification are intended for use in 7.62mm weapons.

6.2 Acquisition requirements. Acquisition documents must specify the following.

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Type and level of packing for the cartridge.

d. Provisions for the submission of Inspection Equipment Designs, see 6.3.

e. Provisions for submission of first article sample.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. Request letter of submittal shall state the contractor, contract number, specification number, item nomenclature and classification of defects or tests. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Hazard notice. The cartridge described herein and certain of its components are flammable and explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

6.5 Drawings. Drawing listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARDC, ARRADCOM, Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.6. Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.7 Subject term (key word) listing.

Ammunition
Bullet Integrity
Function and Casualty Tests
Piezoelectric

6.8 Shipping. When cartridges from more than one lot are shipped at one time, each lot shall be kept separate and the division between the lots clearly indicated to prevent mixing of the lots in transit.

6.9 International agreements. Certain provisions of this specification are the subject of international standardization agreements listed below. When amendment, revision or cancellation of this specification is proposed which will affect or violate the International Agreement concerned, the preparing activity will take appropriate reconcilliation action through international standardization channels including Departmental Standardization Offices to change the agreement or make other appropriate accommodations.

NATO STANAG	ABCA-ARMY-STD
2310	123 2315
2316	
2320	

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6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1305-AD79)

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APPENDIX A

DIM TRACE TEST PROCEDURE

10. SCOPE

10.1 Scope. This appendix details the procedure to determine the relative effectiveness of dim tracer ammunition and disclose the path of bullet flight by firing it under controlled conditions. The acceptance or rejection is based on the criteria specified herein. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. DEFINITION OF TERMS

30.1 "Full Luminosity": is defined as any trace near or at full luminosity as observed by the naked eye, and occurring for at least 100 yards during the length of the trajectory. Such trace is considered unacceptable.

30.2 "Satisfactory Trace": The trace of the bullet is visible to the observers using the starlight scope per requirements.

30.3 "Blind Trace": No visible trace using the starlight scope, during the entire trajectory.

30.4 "Short Trace": Trace did not reach required length, but no other defect.

30.5 "Delayed": Trace starts late and traces the required distance.

30.6 "Partial": Trace starts late and does not trace the required distance.

30.7 "Other": Any other than listed above.

40. EQUIPMENT

40.1 Equipment listed in trace test section of the appropriate Inspection Equipment List shall be used.

50. OBSERVATION POINTS

50.1 Three observers shall be used, two using starlight scopes and one using naked eye.

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50.1.1 One starlight scope shall be positioned behind and approximately 20 feet to the side of the weapon, so as to adequately observe trace performance to include as much of the trajectory from muzzle to 850 yards as possible.

50.1.2 The naked eye observer will assume a position similar to 50.1.1 above.

50.1.3 The other starlight scope shall be positioned down range not less than 400 yards from the muzzle so as to accurately and consistently determine trace to a point not less than 850 yards from the muzzle.

50.1.4 Observation at weapon shall be made for any abnormal behavior as an informational check only. Record shall be made of erratic flight, bursting bullets, muzzle flash of obviously greater intensity than that produced by other bullets in the sample, or such other behavior as may be considered abnormal by the observer.

60. TEST PROCEDURE

60.1 Preparation for Firing

60.1.1 All trace testing shall be performed at night. All lights interfering with starlight scope observations will be turned off.

60.1.2 Weapons shall conform to the requirements shown on the applicable drawings. To assure that a barrel used in the Trace test is within the serviceable life, it shall be used for the Trace test only, and shall be disqualified after firing 6000 cartridges.

60.1.3 The weapon is assembled in the fixture on the mount. The pier or mount shall be of solid construction.

60.1.4 Ammunition shall be assembled in the appropriate links.

60.1.5 The ammunition shall not be conditioned to temperature but shall be fired under existing conditions; however, to prevent temperature of weapon from affecting cartridge performance, if any unusual delay in firing occurs after the cartridge is chambered, such cartridge shall be extracted and the next cartridge in sequence chambered and fired.

60.1.6 The test barrel shall be at ambient temperature prior to firing each test.

60.2 Firing the Test

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60.2.1 At least three unrecorded Cartridges of the type of ammunition under test shall be fired to sight, warm and foul the weapon prior to each test.

60.2.2 The cartridges shall be fired single-shot in a uniform sequence, with sufficient interval between cartridges to allow the trace of each individual shot to be observed and recorded by each observer.

70. CLASSIFICATION OF DEFECTS

70.1 The following are considered defects and will count against trace performance:

a. Full Luminosity

b. Blind Trace

c. Short Trace

d. Partial Trace

70.2 The following are not considered defects but will be reported during performance testing:

a. Delayed Trace

80. RECORDING OF DATA

80.1 The trace characteristics of each shot fired are recorded by each observer. After the test is completed the observers must check their respective observations together shot by shot. Similar defects reported by each observer shall be recorded as one failure.

80.2 The test results are reported as follows:

a. Number of cartridges fired.

b. Satisfactory trace (percentage).

c. Number and type of defects.

d. Number and type of case casualtied.

80.3 The following test weapon data shall be recorded on the test sheet:

a. Receiver number

b. Barrel number

c. Total number of cartridges fired in barrel (prior to test).

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APPENDIX B

BULLET INTEGRITY TEST PROCEDURE

10. SCOPE

10.1 Scope. This appendix details the procedure for bullet integrity test. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. EQUIPMENT

30.1 Equipment listed in the Trace section of the Inspection Equipment List shall be used.

30.2 Target screen 6' x 6' shall be placed 50 feet from the muzzle of the gun, in the line of fire.

40. OBSERVATION POINTS

40.1 The three observers used for the Trace test shall also observe for the Bullet Integrity Test.

50. TEST PROCEDURE

50.1 Preparation for Firing.-

50.1.1 Preparation for firing shall be in accordance with SCATP-7.62, Chapter 1, Section 17.

50.1.2 Prior to each test, the gun to be used for the test shall be inspected. Gun shall be in clean condition, chamber and bore of the barrel shall also be clean.

50.1.3 Target screen, as referenced in SCATP-7.62, Chapter 1, Section 15, paragraph 15.3.2, shall be placed 50 from the muzzle for the gun, so that the bullets pass through the screen at its center.

50.2 Firing the Test.-

50.2.1 Test shall be fired in accordance with SCATP-7.62.

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50.2.2 Each observer shall keep a shot-by-shot record. In addition to recording trace performance of each bullet, observers shall also record any abnormalities observed, e.g., large flash, loud report, erratic flight, absence of noise of passing bullet at terminal point, etc. Observers shall record performance of each bullet in order to compare records upon completion of the test.

50.2.3 The target screen shall be examined after each series of 20 rounds or anytime a suspect bursting or stripped bullet is encountered. Screen shall be renewed as often as is necessary to assure positive identification of bursting or stripped bullets.

60. RECORDING OF DATA

60.1 Upon completion of the test, the observers shall check their respective observations together, shot-by-shot. All observations shall be reported.

60.1.1 Suspect shots shall be recorded by each observer, as follows:

a. Loud Report
b. Large Flash
c. Lack of Trace
d. Erratic Flight
e. Shot Number

60.1.2 Target screen observations shall be reported as follows:

a. Number of Holes
b. Location of Holes
c. Type of Holes

60.1.3 The test results are reported as follows:

a. Number of Cartridges Fired
b. Number and Type of Defects
c. Number and Type of Case Casualties

60.1.4 The following test weapon data shall be recorded on the test sheet:

a. Receiver Number
b. Barrel Number
c. Total Number of Cartridges Fired in Barrel prior to Test.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-50704A (AR)	20 SEPTEMBER 1991

3. DOCUMENT TITLE

CARTRIDGE, 7.62MM: DIM TRACER, XM276

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

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